ACICP302ICS – Network Security & Attacks

OT Security Simulation Using Modbus/TCP

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Abstract

This laboratory exercise investigated security monitoring and attack detection mechanisms within a simulated Operational Technology (OT) environment. The supplied OT Security Simulation was deployed inside an isolated Ubuntu Virtual Machine with all communications restricted to the localhost interface (127.0.0.1). The experiment examined the behavior of Modbus/TCP communications under normal operating conditions and during simulated cyber-attacks. Several controlled attack scenarios were executed against both the direct communication path and a guarded communication path protected by validation rules. Results were collected through dashboards, packet captures, system logs, and event records. The findings demonstrated that rule-based validation significantly improved detection and mitigation of abnormal process commands while highlighting the limitations of relying solely on threshold-based security controls. The laboratory provided practical understanding of OT security concepts, attack techniques, and defensive mechanisms within industrial control systems.

1. Introduction

Operational Technology (OT) systems are responsible for monitoring and controlling physical industrial processes. These systems commonly use industrial communication protocols such as Modbus/TCP to exchange information between field devices, controllers, and monitoring systems. Due to increasing connectivity between Information Technology (IT) and OT environments, industrial systems have become attractive targets for cyber attackers.

The purpose of this laboratory exercise was to examine how abnormal commands and malicious process manipulations can affect industrial systems and to evaluate the effectiveness of security validation controls implemented within a simulated environment. The exercise utilized the instructor-supplied OT Security Simulation running entirely within an isolated virtual machine to ensure safe experimentation.

2. Objectives

The objectives of this laboratory exercise were:

1. To deploy and operate the supplied OT security simulation.
2. To analyze Modbus/TCP communications in a controlled environment.
3. To observe normal process behavior and collect baseline measurements.

4. To perform controlled attack demonstrations against the simulated plant.
5. To compare the behavior of direct and guarded communication paths.
6. To evaluate the effectiveness of rule-based validation controls.
7. To collect and analyze packet captures, logs, and event records.
8. To assess the strengths and limitations of OT security controls.

3. Laboratory Environment

3.1 System Configuration

Component	Specification
Operating System	Ubuntu LTS
Virtualization Platform	VMware Workstation / VirtualBox
Python Environment	Python 3 Virtual Environment
Communication Protocol	Modbus/TCP
Network Scope	Localhost Only (127.0.0.1)

Simulated Components

Component	Address
Plant Simulator	127.0.0.1:5020
Validation Guard	127.0.0.1:5021
Direct Dashboard	http://127.0.0.1:8080
Guarded Dashboard	http://127.0.0.1:8081

3.2 Safety Verification

Prior to running the simulation, verification checks were performed to ensure compliance with laboratory safety requirements.

Unsafe Address Check

Terminal

grep -R -nE '192\.168\.2\.139|0\.0\.0\.0' plant_sim

Result: No unsafe network addresses were detected.

Loopback Verification

Terminal

grep -R -n '127\.0\.0\.1' plant_sim

Result: All services were configured for localhost communication.

Network Isolation

The VM network adapter remained disconnected during experimentation. No bridged networking, port forwarding, or external network communication was used throughout the laboratory exercise.

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4. Baseline System Observations

After starting the plant simulator, validation guard, direct dashboard, and guarded dashboard, baseline process values were recorded.

Initial Process Readings

Process Variable	**Observed Value**
Temperature	Approximately 67°C
Pressure	Approximately 34 bar
Flow Rate	Approximately 90 Nm^3/h

Normal System Behavior

- All services started successfully.
- Dashboard values refreshed continuously.
- Both dashboards displayed consistent information.

- No warning or alert messages were generated.
- System logs indicated normal plant operation.

These observations established the baseline operating condition before attack demonstrations were performed.

5. Packet Capture Analysis

A packet capture was conducted using Wireshark on the Linux loopback interface (lo).

Observations

The capture showed continuous Modbus/TCP communication between the dashboards, guard service, and plant simulator.

A typical communication sequence consisted of:

1. Client request.
2. Modbus register access.
3. Plant response.
4. Dashboard update.

Analysis

The packet capture confirmed that:

- All communication remained on the localhost interface.
- No external network activity occurred.
- Modbus requests and responses followed expected protocol behavior.
- Register values corresponded with dashboard readings.

The packet analysis demonstrated how industrial devices exchange process information using Modbus/TCP.

6. Attack Demonstrations

Three attack demonstrations were executed using the supplied scripts without modification.

6.1 Attack Hold Demonstration

Commands Executed

Terminal

python3 attack_hold.py 5020

Terminal

python3 attack_hold.py 5021

Objective

To attempt writing a pressure value outside the normal operating range.

Observations

Direct Path (Port 5020)

- The abnormal pressure value was accepted.
- Dashboard readings reflected the manipulated value.
- No validation controls prevented the action.

Guarded Path (Port 5021)

- The validation guard detected the abnormal request.
- The value was rejected.
- Event logs recorded the attempted violation.

Analysis

The experiment demonstrated that systems accepting unrestricted input are vulnerable to process manipulation. The guard successfully enforced predefined operational limits and prevented invalid commands from reaching the plant.

Conclusion

Range validation was effective in blocking obviously dangerous process values.

6.2 False Data Attack

Commands Executed

Terminal

python3 false_data_attack.py 5020

Terminal

python3 false_data_attack.py 5021

Objective

To inject unrealistic process values into the simulated environment.

Observations

Direct Path

- Process values changed significantly.
- Dashboards displayed manipulated measurements.
- The attack altered perceived process conditions.

Guarded Path

- Suspicious values triggered validation checks.
- Multiple abnormal entries were rejected.
- Events were recorded in the system log.

Analysis

False data injection attacks can create misleading process conditions and affect operator decision-making. The validation guard reduced this risk by enforcing acceptable operating bounds.

Conclusion

The guard improved detection and mitigation of abnormal process measurements.

6.3 Slow Drift Attack

Commands Executed

Terminal

python3 slow_drift_attack.py 5020

Terminal

python3 slow_drift_attack.py 5021

Objective

To gradually alter process values over time in an attempt to avoid traditional detection methods.

Observations

Direct Path

- Gradual pressure changes accumulated successfully.
- Dashboard values slowly deviated from normal conditions.

Guarded Path

- The system generated warnings when drift thresholds were exceeded.
- Some requests were rejected after detecting unusual trends.

Analysis

Slow drift attacks are more challenging to identify because individual changes appear legitimate. The guard attempted to monitor cumulative deviations rather than relying solely on absolute thresholds.

Conclusion

Behavior-based validation provides additional protection against gradual manipulation attacks.

7. Comparison of Direct and Guarded Architectures

Security Feature	Direct Path	Guarded Path
Input Validation	No	Yes
Range Enforcement	No	Yes
Event Logging	Limited	Enhanced
Detection Capability	Low	Moderate
Attack Resistance	Weak	Improved
Operational Visibility	Basic	Enhanced

The guarded architecture consistently provided stronger protection than the direct communication path.

8. Security Evaluation

Strengths

The laboratory environment demonstrated several effective security measures:

- Localhost-only communication reduced attack exposure.
- Validation checks prevented many abnormal writes.
- Event logging provided useful forensic information.
- Isolated virtual machine operation ensured safe experimentation.
- Security controls improved visibility into suspicious activity.

Weaknesses

Several limitations were identified during testing:

- No authentication mechanisms were implemented.
- No user authorization controls existed.
- Direct communication bypass remained possible.

- Validation relied primarily on predefined rules.
- More sophisticated attacks could potentially evade detection.
- The simulation does not represent a complete industrial safety system.

Risk Assessment

Threat	Likelihood	Impact	Risk Level
Unauthorized Process Changes	Medium	High	High
False Data Injection	High	Medium	High
Replay Attack	Medium	Medium	Medium
Slow Drift Manipulation	Medium	Medium	Medium
Denial of Service	Low	High	Medium

9. Learning Outcomes

This laboratory exercise provided practical experience in:

- Industrial Control System security principles.
- Modbus/TCP protocol analysis.
- Network packet capture techniques.
- Security monitoring and event analysis.
- Attack simulation methodologies.
- Validation-based defense mechanisms.
- OT risk assessment practices.
- Evidence collection and documentation.

The exercise highlighted the importance of layered security controls in industrial environments and demonstrated how monitoring, validation, and logging can improve system resilience.

10. Conclusion

The OT Security Simulation successfully demonstrated the effects of cyber-attacks on industrial control environments and the benefits of implementing validation-based security mechanisms. The comparison between direct and guarded communication paths showed that the validation guard effectively detected and blocked several categories of abnormal requests, including out-of-range values and suspicious process manipulations.

The laboratory also revealed important limitations. While rule-based validation improved security visibility and reduced risk, it did not provide comprehensive protection against all attack scenarios. Effective industrial cybersecurity requires multiple defensive layers including authentication, authorization, monitoring, network segmentation, secure configurations, and appropriate operational procedures.

Overall, the experiment achieved all learning objectives and provided valuable insight into OT system vulnerabilities, attack detection techniques, and defensive security controls within industrial environments.

References

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